

31 January 2023

December 2022 Quarterly Activities Report

- Drilling at **Murchison Gold Project (MEK 100%)** continued to deliver strong results at St Anne's, outside the 1.2Moz gold Mineral Resource, including:
 - **4m @ 87.50g/t Au** from 172m (22SARC005)
 - **16m @ 3.48g/t Au** from 48m including **8m @ 6.17g/t Au** (22SAAC165)
 - **12m @ 2.48g/t Au** from 76m including **8m @ 3.55g/t Au** (22SAAC164)
 - **12m @ 2.33g/t Au** from 44m including **4m @ 3.96g/t Au** (22SAAC152)
 - **12m @ 1.38g/t Au** from 88m including **4m @ 2.27g/t Au** (22SAAC128)
 - **12m @ 1.13g/t Au** from 80m (22SAAC131)
- Shallow aircore drilling also intersected broad gold zones outside the Mineral Resource on the Western Flank of Turnberry (685,000oz @ 2.0g/t Au), part of the Murchison Gold Project, results included:
 - **12m @ 2.01g/t Au** from 20m including **4m @ 5.23g/t Au** (22TBAC054)
 - **12m @ 1.70g/t Au** from 16m including **4m @ 3.30g/t Au** (22TBAC064) and **4m @ 1.23g/t Au** from 56m (22TBAC064)
 - **16m @ 1.18g/t Au** from 96m including **4m @ 3.10g/t Au** (22TBAC042)
 - **4m @ 3.31g/t Au** from 32m (22TBAC041)
 - **8m @ 2.09g/t Au** from 36m including **4m @ 3.78g/t Au** (22TBAC050)
- Metallurgical test results were received for St Anne's confirming gold recoveries averaging 98%.
- Diamond drilling was completed at **Circle Valley (MEK 100%)** targeting structural information prior to a larger drill program in January 2023, following up on 2022 drill results, including:
 - **23m @ 5.09g/t Au** from 13m including **10m @ 9.35g/t Au** (22CVRC001)
 - **16m @ 3.06g/t Au** from 32m including **4m @ 10.80g/t Au** (22CVRC009)
 - **10m @ 4.72g/t Au** from 120m including **4m @ 10.20g/t Au** (22CVRC008)
 - **32m @ 1.39g/t Au** from 4m including **8m @ 4.96g/t Au** (22CVRC025)
- Circle Valley rare earth metallurgical test work advanced significantly with upstream beneficiation tests showing up to 90.4% of the valuable rare earth elements can be recovered in the -20µm size fraction indicating grade can be increased by up to 91% using low cost and simple beneficiation processes.
- The Company's strong cash position of \$6.7m at quarter end allows for continued aggressive drilling activities across the project portfolio.

Commenting on the quarter, Meeka's Managing Director Tim Davidson said: "We completed another active quarter with teams advancing our Murchison and Albany-Fraser projects. The bulk of our work focussed on the Murchison Gold Project where we drilled over 42,000m safely and productively to deliver Mineral Resource growth. Based on this drilling we upgraded the Turnberry Mineral Resource in January 2023, and will deliver an initial Mineral Resource for St Anne's in the March 2023 quarter.

Diamond drilling was also completed at Circle Valley to inform a geology and structural model to guide extensional drilling in 2023, following up the strong gold results received during 2022.

Also at Circle Valley, positive results were returned from our ongoing rare earth metallurgical test program. The results show that the mineralisation can be upgraded using low cost and simple beneficiation processes prior to metal extraction through leaching. The positive results provide confidence to advance our rare earth projects to the Mineral Resource stage, for which drilling commenced in January 2023.

Finally, we ended the quarter with a strong cash position of \$6.7 million, which will be actively deployed with ongoing drilling, metallurgical testing and feasibility study work providing strong news through the year.”

Meeka Metals Limited (“**Meeka**” or “**the Company**”) is pleased to provide a summary of activities completed during the December 2022 quarter.

The Company was active in advancing its gold and rare earths projects with work ongoing throughout the quarter in the Murchison, and at Circle Valley during October 2022 where an additional field team mobilised to that project.

Drilling continued at the Murchison Gold Project, focussed on Mineral Resource growth. New zones of shallow oxide gold were drilled at St Anne's. Furthermore, RC drilling intersected high grade primary mineralisation with visible gold below previously identified zones of oxide mineralisation.

At Turnberry, shallow oxide gold was intersected outside the Mineral Resource and confirms the continuity of mineralisation to the north, into a previously untested zone. Drilling also targeted a conceptual southern extension to the Western Flank mineralisation and this also returned shallow oxide gold. Follow up drilling will commence in 2023.

Diamond drilling at Circle Valley was completed in October 2022. Four holes were drilled delivering valuable information in advance of extensional drilling in the March 2023 quarter.

Positive rare earth metallurgical test results from Circle Valley were received in November 2022, which in addition to development of a Mineral Resource, support future economic studies.

The Company finished the quarter strongly with a cash position of \$6.7m.

Murchison Gold Project

Drilling continued in the Murchison throughout the quarter, focussed on Mineral Resource growth at Turnberry and St Anne's, which is to be included in a Mineral Resource in the March 2023 quarter.

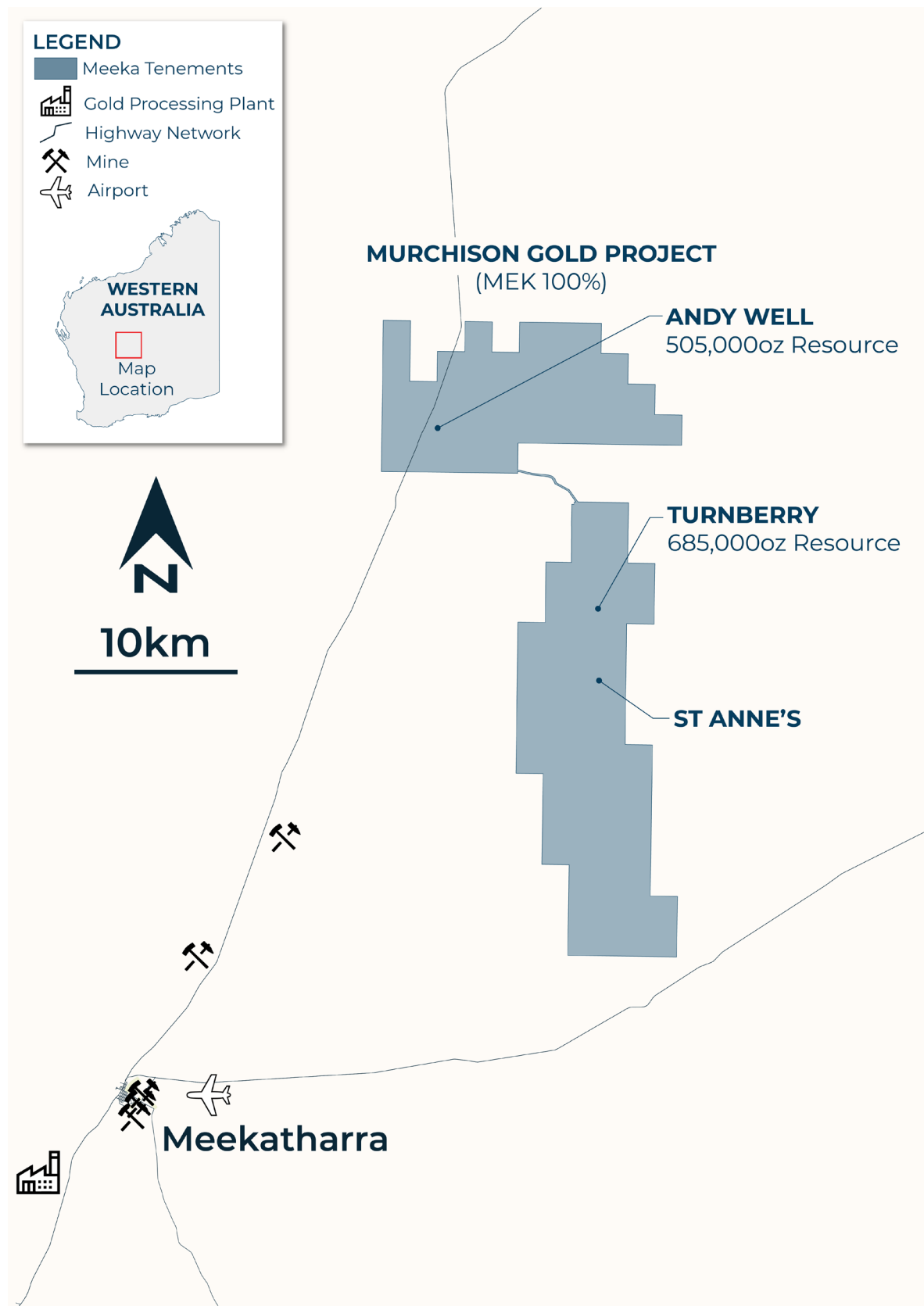


Figure 1: Andy Well, Turnberry and St Anne's locations.

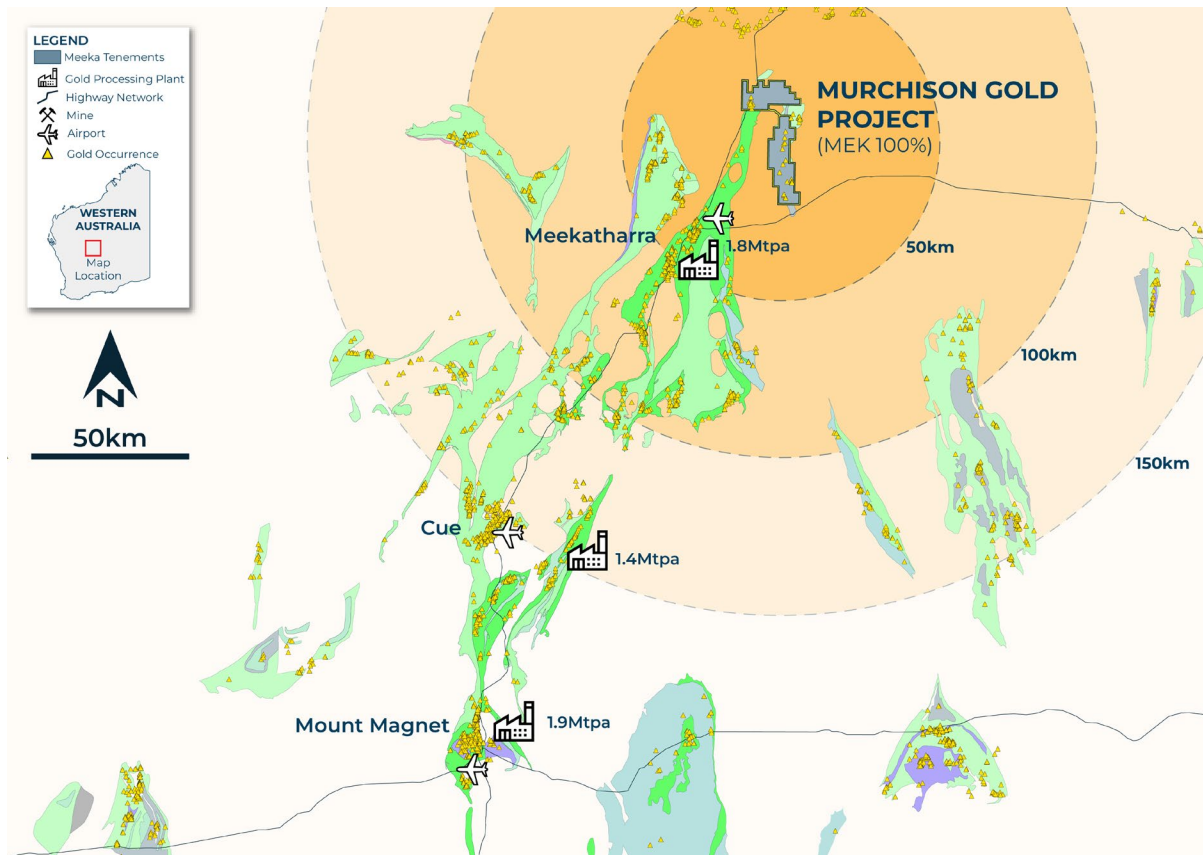


Figure 2: Meeka's 100% owned Murchison Gold Project location.

Shallow oxide gold was intersected on the Western Flank of Turnberry, outside the Mineral Resource (685,000oz @ 2.0g/t Au), and confirm the continuity of mineralisation to the north and south, into a previously untested zone.

New assays from the northern extension to the shallow Western Flank mineralisation include:

- **12m @ 2.01g/t Au** from 20m including **4m @ 5.23g/t Au** (22TBAC054)
- **12m @ 1.70g/t Au** from 16m including **4m @ 3.30g/t Au** (22TBAC064) and **4m @ 1.23g/t Au** from 56m (22TBAC064)
- **8m @ 2.09g/t Au** from 36m including **4m @ 3.78g/t Au** (22TBAC050) and **12m @ 0.67g/t Au** from 56m (22TBAC050)
- **8m @ 1.05g/t Au** from 52m (22TBAC065)

New assays from the southern extension to the shallow Western Flank mineralisation include:

- **16m @ 1.18g/t Au** from 96m including **4m @ 3.10g/t Au** (22TBAC042)
- **4m @ 3.31g/t Au** from 32m (22TBAC041) and **16m @ 0.48g/t Au** from 60m (22TBAC041)
- **20m @ 0.63g/t Au** from 40m including **4m @ 1.34g/t Au** (22TBAC043) and **4m @ 0.33g/t Au** from 112m (22TBAC043)

This shallow, high-grade gold sits on the limb of a folded magnetic quartz dolerite unit, which forms a discrete magnetic anomaly. Aeromagnetic data indicates that the magnetic anomaly extends to the north-northwest of Turnberry, making this area one of a number of exciting extensional targets.

Drilling also tested for southern extensions of Turnberry. Pleasingly, this drilling intersected sediments below a paleo feature, not the magnetic quartz dolerite unit, which principally hosts the gold at Turnberry. Further drilling is planned to define the magnetic quartz dolerite unit.

New assay results from the drilling to the south of Turnberry included:

- **8m @ 1.93g/t Au** from 80m including **4m @ 3.46g/t Au** (22TBAC034)
- **4m @ 0.39g/t Au** from 56m (22TBAC027) and
4m @ 0.33g/t Au from 64m (22TBAC027) and
4m @ 0.53g/t Au from 76m (22TBAC027)
- **4m @ 0.48g/t Au** from 104m (22TBAC028)
- **4m @ 0.78g/t Au** from 72m (22TBAC030)
- **12m @ 0.45g/t Au** from 76m (22TBAC031)
- **4m @ 0.31g/t Au** from 72m (22TBAC039) and
4m @ 0.87g/t Au from 80m (22TBAC039)

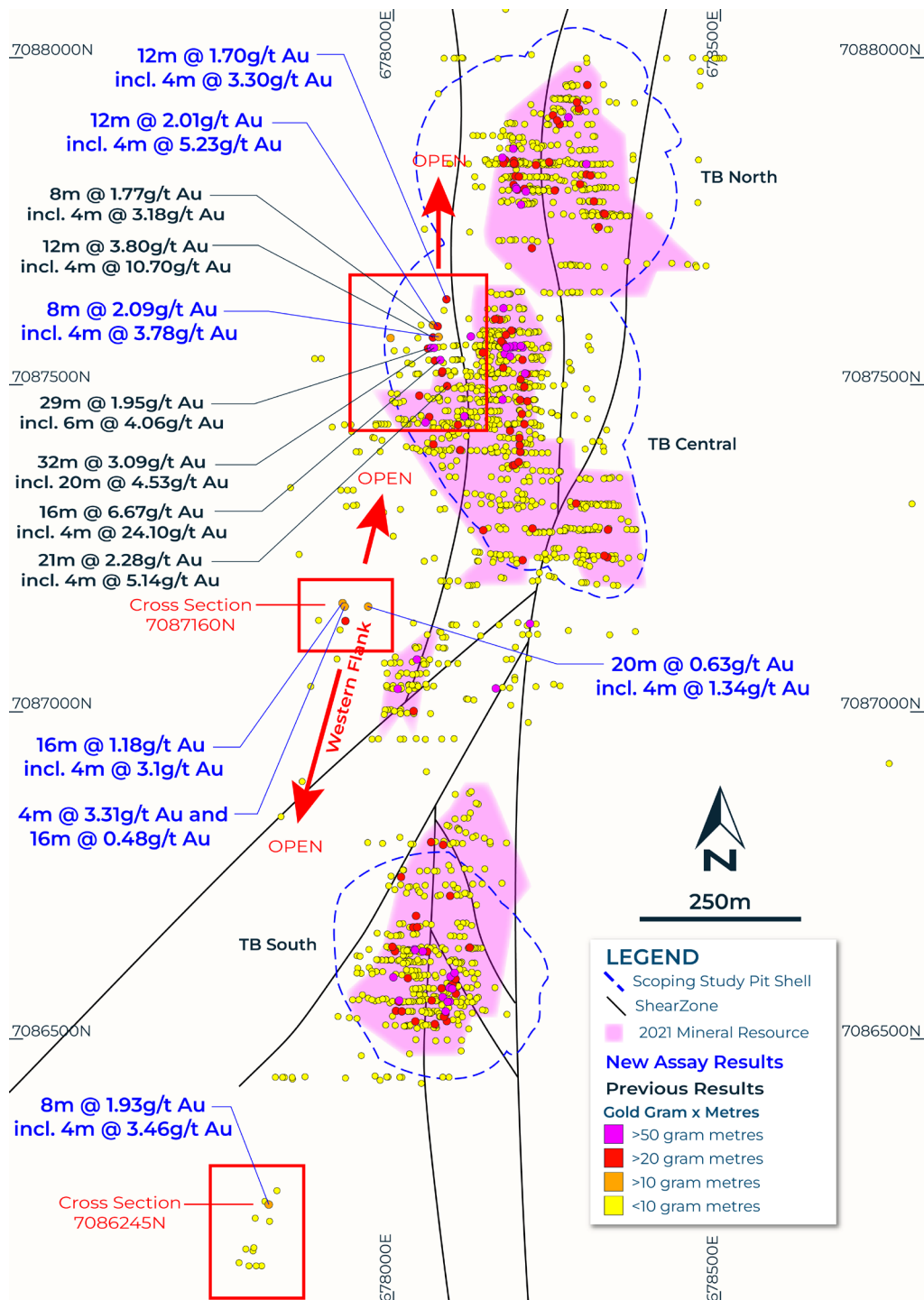


Figure 3: Plan showing new Turnberry assays. Position of cross sections (Figure 4 & 5) is shown in red text. Turnberry remains open along strike and at depth.

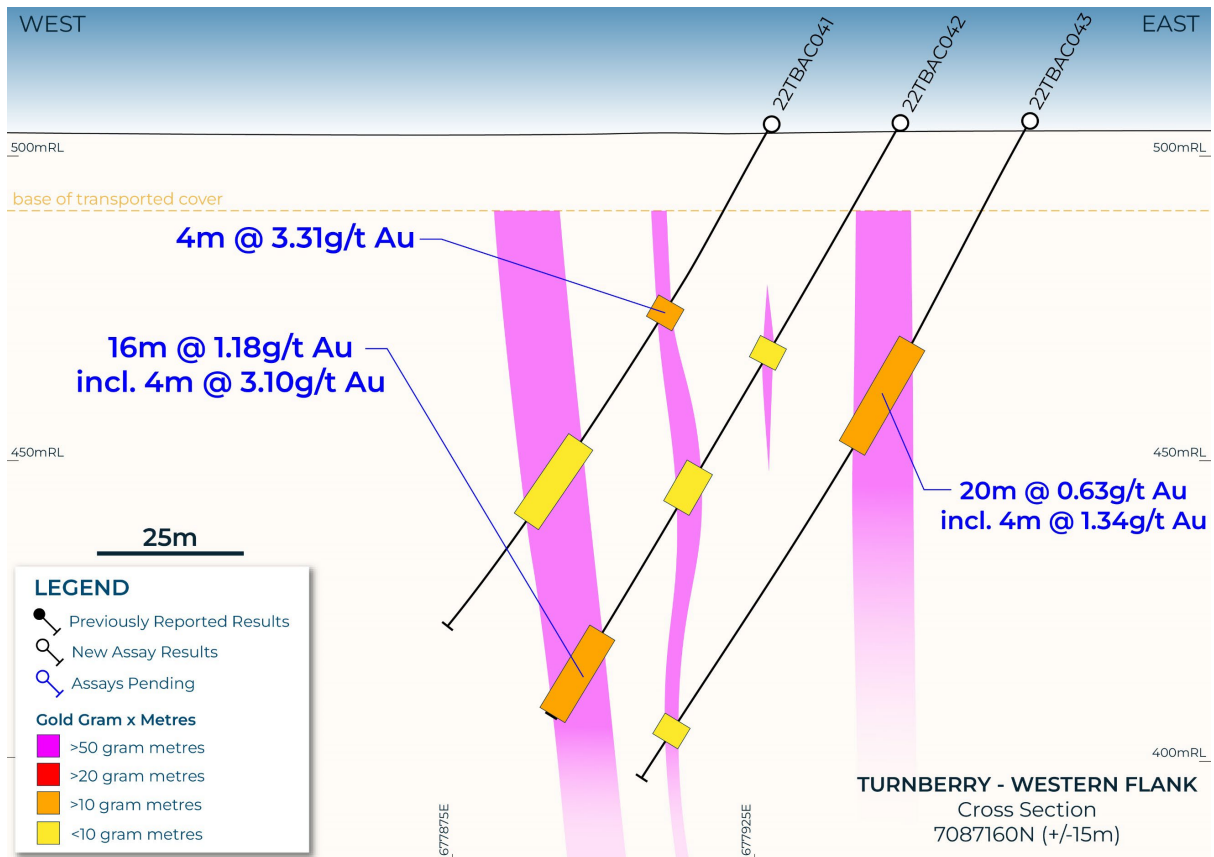


Figure 4: Cross section 7087160N through the southern extension of Western Flank shallow, oxide gold.

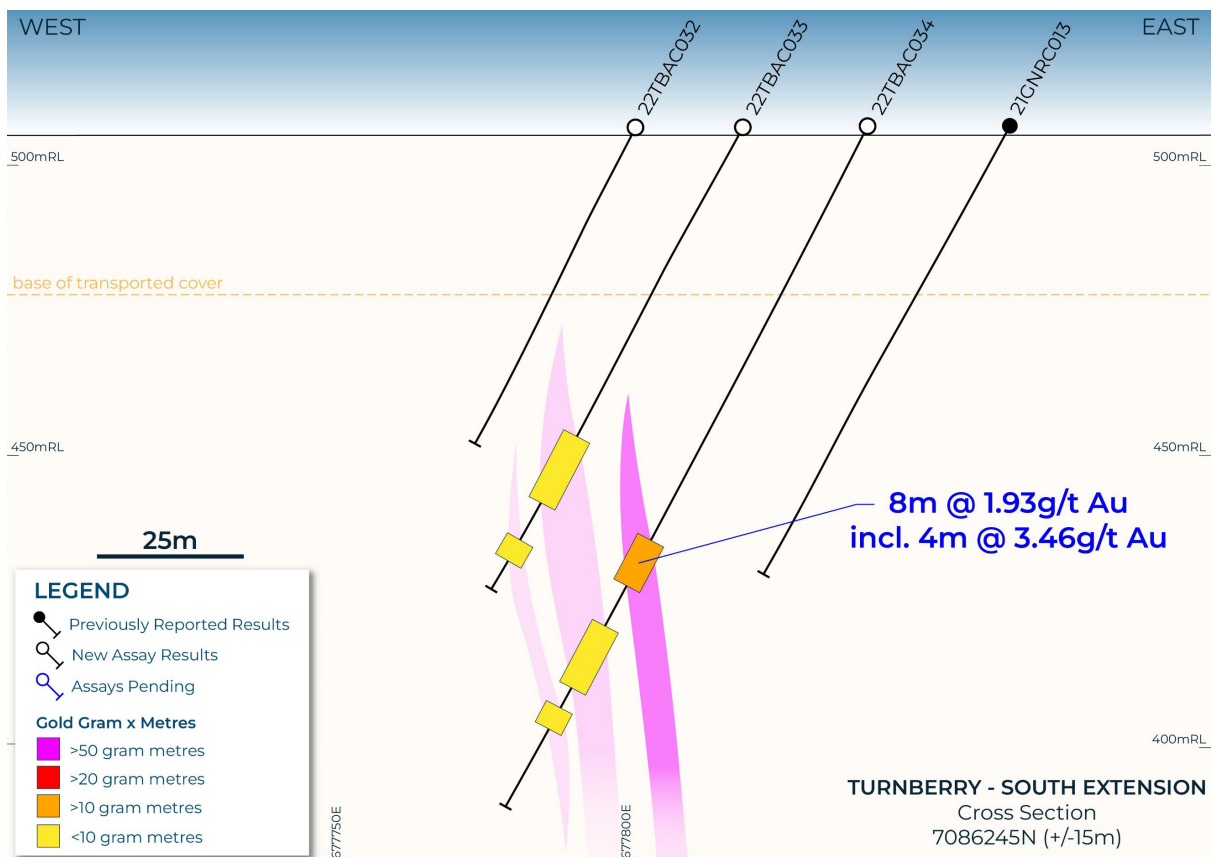


Figure 5: Cross section 7086245N through the southern extension of Turnberry.

Drilling at St Anne's targeting the broad, sub-vertical north-south trending shear zone within mafic rocks, host to the Driver lode, delivered strong gold mineralisation and visible gold. Assays included:

- **4m @ 87.50g/t Au** from 172m (22SARC005)
- **8m @ 1.22g/t Au** from 124m (22SARC001)
- **4m @ 1.54g/t Au** from 128m (22SARC003)

Shallow air core drilling at St Anne's, targeting the previously untested eastern mafic contact, also intersected strong gold mineralisation now called the 'Wood' lode. Assays included:

- **16m @ 3.48g/t Au** from 48m including **8m @ 6.17g/t Au** (22SAAC165)
- **12m @ 2.48g/t Au** from 76m including **8m @ 3.55g/t Au** (22SAAC164)
- **12m @ 2.33g/t Au** from 44m including **4m @ 3.96g/t Au** (22SAAC152)
- **12m @ 1.38g/t Au** from 88m including **4m @ 2.27g/t Au** (22SAAC128)
- **12m @ 1.13g/t Au** from 80m (22SAAC131)

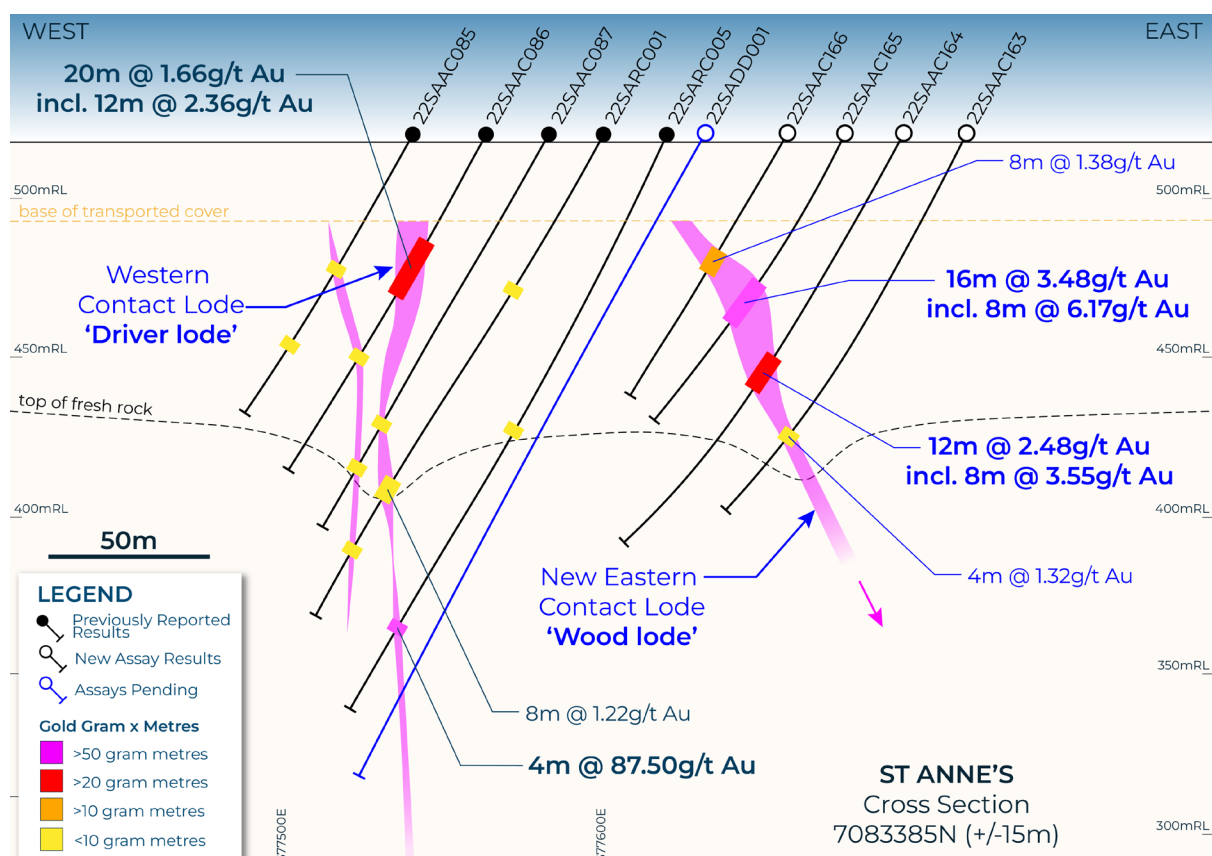


Figure 6: Cross section 7083385N through St Anne's showing the new eastern contact 'Wood' lode.

Circle Valley – Gold

Diamond drilling at Circle Valley was completed in October 2022. The drilling intersected a high-grade metamorphic assemblage intruded by late narrow pegmatitic intrusions. Silica flooding and potassic alteration with biotitic zones is observed in the core, as well as potassic feldspar associated with zones of migmatisation, including visual indications of gold mineralisation. These holes will provide valuable information in advance of extensional drilling in the March 2023 quarter.

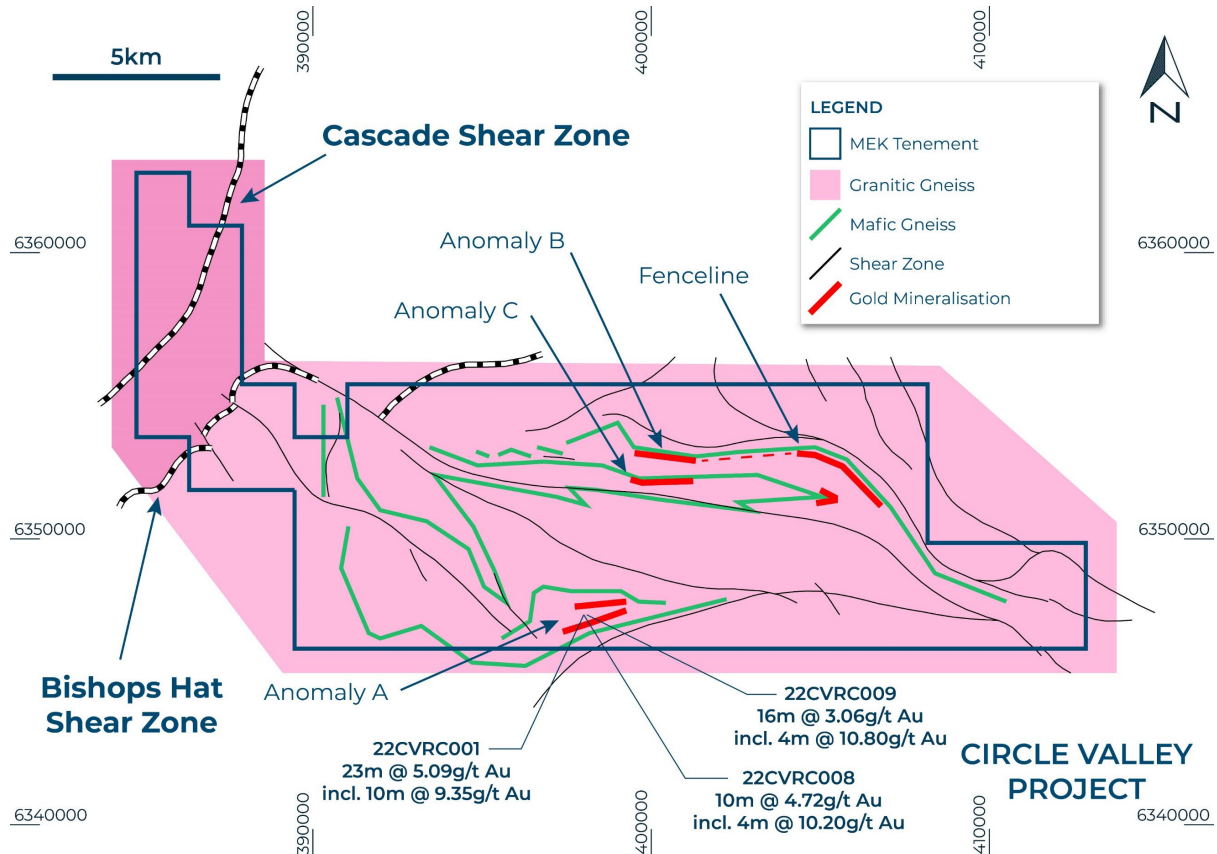


Figure 7: Meeka's 100% owned Circle Valley Project showing location of Anomaly A where diamond drilling was completed.

Circle Valley – Rare Earths

Following the initial rare earth metallurgical test program completed by Australian Nuclear Science and Technology Organisation ("ANSTO") in July 2022, further positive upstream beneficiation test results were received in November 2022. The results confirm that low cost, simple and broadly adopted large scale mineral processing techniques can be applied to remove waste from the process stream prior to leaching. On a commercial scale this would involve particle size separation through scrubbing to produce a slurry prior to screening and hydro-cyclone classification to discard the non-target elements in the coarse fraction, while retaining the valuable -20µm size fraction, which hosts most of the rare earth elements.

Key results from the beneficiation test work were:

- Between 90.4% and 62.5% of the rare earth elements were contained in the fine fraction (-20µm size fraction).
- By rejecting the coarse fraction (+20µm), between 65.5% and 39.6% of the sample mass can be removed while retaining most of the rare earth elements.
- Rejecting the coarse fraction (+20µm) delivered TREO grade uplift of between 1.15 times and 1.91 times compared to the raw sample grade.

These results, in addition to development of a rare earth Mineral Resource, will support future economic studies.

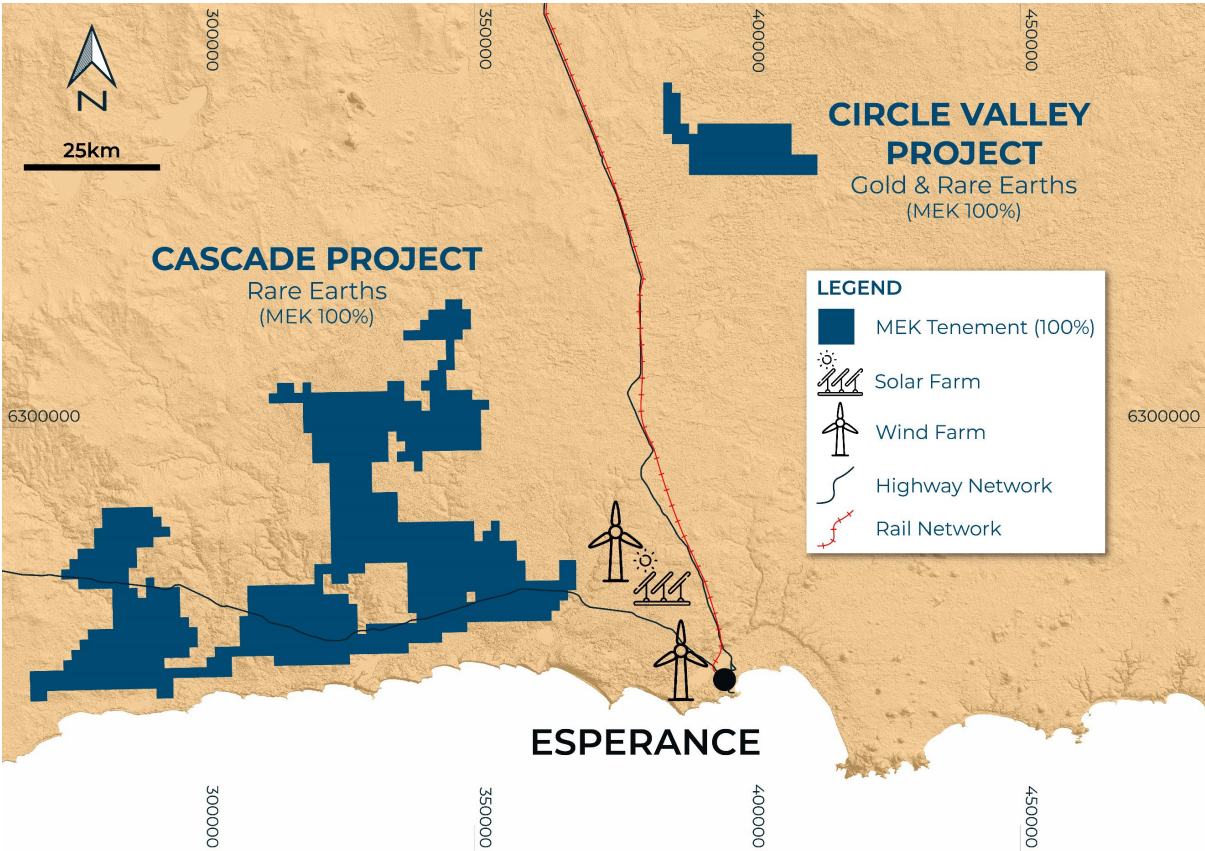


Figure 8: Plan showing the Company’s 100% owned Albany-Fraser projects.

CORPORATE

Financial

An Appendix 5B – Quarterly Cash Flow Report for the quarter ended 31 December 2022, accompanies this Activities Report.

During the quarter the Company spent approximately:

- \$1.8m on exploration and evaluation activities.
- \$283k on administration and corporate costs.
- \$122k in payments were made to related parties and their associates for director fees, company secretarial fees, bookkeeping fees and legal fees.

The Company ended the quarter with \$6.7m in cash and no debt.

Capital Structure

The capital structure of the Company at 31 December 2022:

Description	Number
Fully Paid Ordinary Shares	1,058,442,264
Unlisted options exercisable at \$0.040 each, expiring 15 Feb 2025	500,000
Unlisted options exercisable at \$0.040 each, expiring 31 Jan 2025	30,000,000
Unlisted options exercisable at \$0.050 each, expiring 25 May 2025	300,000
Unlisted options exercisable at \$0.075 each, expiring 25 May 2025	300,000
Unlisted options exercisable at \$0.100 each, expiring 25 May 2025	600,000
Unlisted options exercisable at \$0.060 each, expiring 4 April 2025	900,000
Unlisted options exercisable at \$0.080 each, expiring 4 April 2025	900,000
Unlisted options exercisable at \$0.100 each, expiring 4 April 2025	1,800,000
Unlisted options exercisable at \$0.060 each, expiring 1 May 2025	875,000
Unlisted options exercisable at \$0.080 each, expiring 1 June 2025	875,000
Unlisted options exercisable at \$0.100 each, expiring 1 July 2025	1,750,000
Performance Rights Class B – \$0.075 VWAP (15-day) price hurdle, expiring 8 August 2023*	34,600,000
Performance Rights Class C – \$0.100 VWAP (15-day) price hurdle, expiring 8 August 2023	21,650,000
Performance Rights Class D – \$0.125 VWAP (20-day) price hurdle, expiring 7 July 2026	38,250,000

**During the September 2022 quarter, performance hurdles for Performance Rights Class B relating to the Andy Well transaction were met. The Company anticipates these shares will be issued in July-August 2023.*

TENEMENT SCHEDULE

Tenements held or under application at 31 December 2022.

Project	State	Tenement	Status	Interest at start of quarter	Interest at end of quarter
Murchison Gold Project	WA	E 51/1596	Granted	100%	100%
		E 51/1217	Granted	100%	100%
		M 51/870	Granted	100%	100%
		E 51/1625	Granted	100%	100%
		E 51/1626	Granted	100%	100%
		E 51/926	Granted	100%	100%
		E 51/927	Granted	100%	100%
		M 51/882	Granted	100%	100%
Circle Valley	WA	E 63/2007	Granted	100%	100%
		E 63/2214	Granted	100%	100%
Cascade	WA	E 63/2173	Granted	100%	100%
		E 63/2217	Granted	100%	100%
		E 74/712	Granted	100%	100%
		E 74/716	Granted	100%	100%
		E 74/721	Granted	100%	100%
		E 74/722	Granted	100%	100%
		E 74/732	Granted	100%	100%
		E 74/735	Granted	100%	100%

FORTHCOMING ANNOUNCEMENTS

February 2023: Presentation – Bell Potter Unearthed 2023 Virtual Conference.

February – March 2023: Gold assays from drilling at St Anne's, Murchison Gold Project.

February – March 2023: Gold assays from Circle Valley (Anomaly A) extensional drilling.

February – March 2023: Rare earth assays from Circle Valley infill drilling.

March 2023: Interim Financial Report – half year to 31 December 2022.

March 2023: Initial Mineral Resource – St Anne's, Murchison Gold Project.

April 2023: March 2023 Quarterly Activities Report.

May 2023: Presentation – RIU Sydney Resources Round-up Conference.

June 2023: Initial Mineral Resource – Circle Valley (rare earths).

June 2023: Pre-feasibility Study for the Murchison Gold Project.

July 2023: June 2023 Quarterly Activities Report.

August 2023: Presentation – Australian Gold Conference, Sydney.

This announcement has been authorised for release by the Company's Board of Directors.

For further information, please contact:

Tim Davidson – Managing Director
+61 8 6388 2700

info@meekametals.com.au

www.meekametals.com.au

ABOUT MEEKA

Meeka Metals Limited is a gold and rare earths company with a portfolio of high quality 100% owned projects across Western Australia.

Gold

Meeka's flagship Murchison Gold Project has a combined 343km² landholding in the prolific Murchison Gold Fields and hosts a large high-grade 1.2Moz JORC Resource. The Company is actively growing these Resources while also progressing toward production. The release of the Murchison Gold Project Scoping Study in December 2021 outlined a robust Project that produces over 420koz of gold¹.

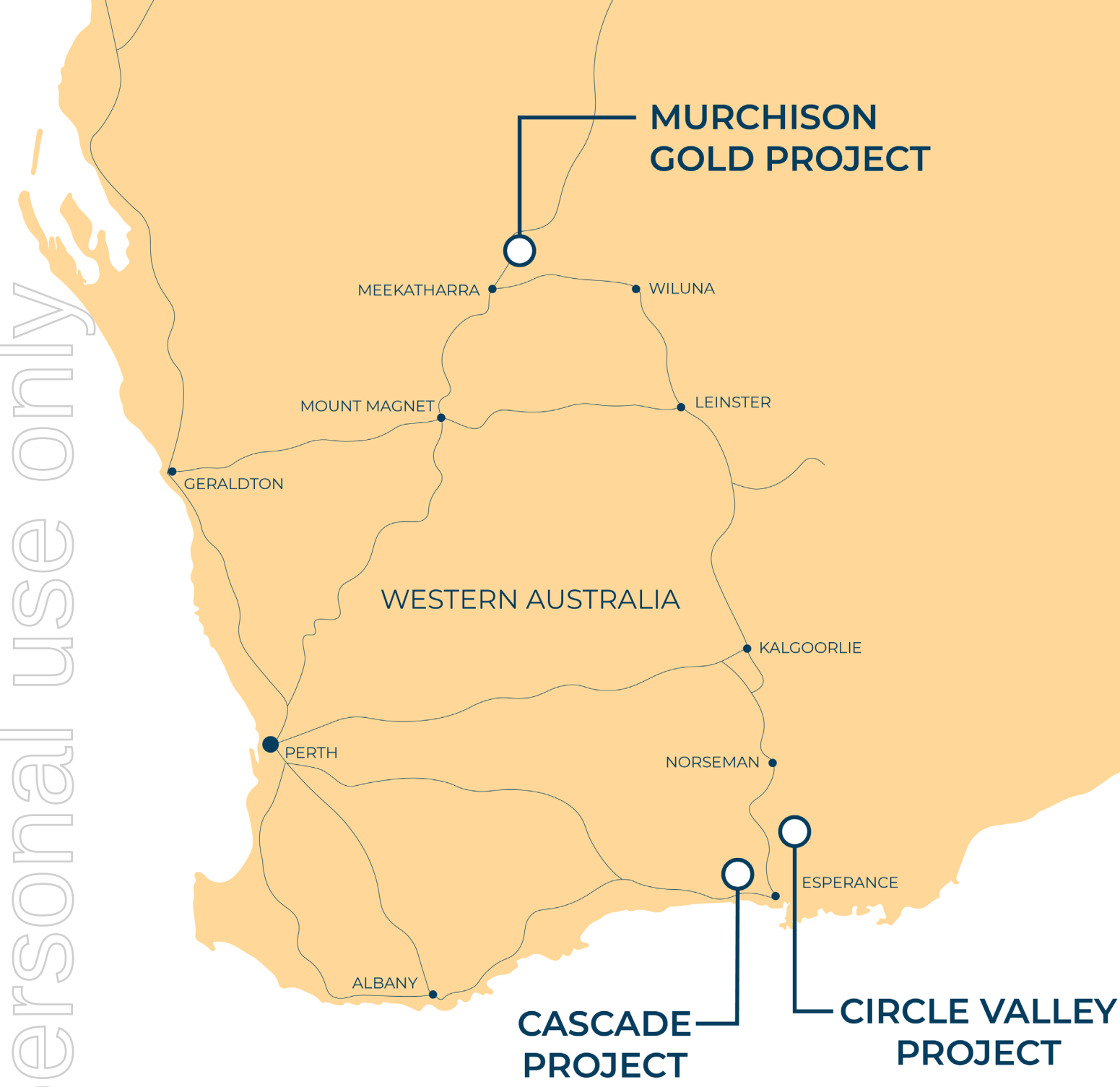
In addition, Meeka owns the Circle Valley Project (222km²) in the Albany-Fraser Mobile Belt (also host to the Tropicana gold mine – 3Moz past production). Gold mineralisation has been identified in four separate locations at Circle Valley and presents an exciting growth opportunity, which is being aggressively pursued.

Rare Earths

Meeka controls the Cascade Rare Earths Project (2,269km²) in a region that is rapidly emerging as a highly prospective clay rare earths province. Importantly, the results to date contain high levels of permanent magnet metals (neodymium-praseodymium). These metals are geopolitically critical, and the Company intends to accelerate its understanding of Cascade through metallurgical work and ongoing drilling.

Circle Valley also hosts clay rare earths within thick, near surface mineralised zones below shallow transported cover. The mineralisation consistently demonstrates a high proportion of the grade as neodymium-praseodymium oxides. Metallurgical work, in addition to infill and extensional drilling are ongoing. An initial Mineral Resource is targeted for 2023.

¹ First reported in ASX announcement dated 1 December 2021 and titled "Murchison Gold Project Scoping Study". The company confirms that it is not aware of any new information or data that materially affects the information included in the original announcement and that all material assumptions and technical parameters underpinning the target continue to apply and have not materially changed. The company confirms that the form and context in which the Competent Person's findings are presented have not been materially modified from the original market announcement.



Mineral Resource Summary

Project	Measured			Indicated			Inferred			Total		
	Tonnes (⁰⁰⁰ t)	Grade (g/t)	Ounces (⁰⁰⁰ oz)	Tonnes (⁰⁰⁰ t)	Grade (g/t)	Ounces (⁰⁰⁰ oz)	Tonnes (⁰⁰⁰ t)	Grade (g/t)	Ounces (⁰⁰⁰ oz)	Tonnes (⁰⁰⁰ t)	Grade (g/t)	Ounces (⁰⁰⁰ oz)
Andy Well	150	11.4	55	1,050	9.3	315	650	6.5	135	1,800	8.6	505
Turnberry				4,600	1.6	230	6,000	2.4	455	10,600	2.0	685
TOTAL	150	11.4	55	5,650	3.0	545	6,650	2.8	590	12,400	3.0	1,190

Notes:

1. Mineral Resources reported to the ASX on 3 January 2023. The company confirms that it is not aware of any new information or data that materially affects the information included in the original announcement and that all material assumptions and technical parameters underpinning the estimates continue to apply and have not materially changed. The company confirms that the form and context in which the Competent Person's findings are presented have not been materially modified from the original market announcement.
2. Mineral Resources are classified in accordance with JORC code (2012).
3. Andy Well Mineral Resource is reported using 0.1g/t cut-off grade.
4. The Turnberry open pit Mineral Resource is only the portion of the Mineral Resource that is constrained within a A\$2,600/oz optimised pit shell and above a 0.5g/t gold cut-off grade.
5. The Turnberry underground Mineral Resource is only the portion of the Mineral Resource that is located outside the A\$2,600/oz optimised pit shell and above a 1.5g/t gold cut-off grade.
6. Estimates are rounded to reflect the level of confidence in the Mineral Resources at the time of reporting.

COMPETENT PERSON'S STATEMENT

The information that relates to Exploration Results as those terms are defined in the 2012 Edition of the "Australasian Code for Reporting of Exploration Results, Mineral Resources and Ore Reserve", is based on information reviewed by Mr Duncan Franey, a Competent Person who is a member of The Australasian Institute of Mining and Metallurgy and the Australian Institute of Geoscientists. Mr Franey is a full-time employee of the Company. Mr Franey has sufficient experience that is relevant to the style of mineralisation and type of deposit under consideration and to the activity being undertaken to qualify as a Competent Person as defined in the 2012 Edition of the 'Australasian Code for Reporting of Exploration Results, Mineral Resources and Ore Reserves'. Mr Franey consents to the inclusion in the report of the matters based on his information in the form and context in which it appears.

The information that relates to Mineral Resources was first reported by the Company in its announcement to the ASX on 3 January 2023. The Company is not aware of any new information or data that materially affects the information included in this announcement and that all material assumptions and technical parameters underpinning the estimates continue to apply and have not materially changed. The company confirms that the form and context in which the Competent Person's findings are presented have not been materially modified from the original market announcement.

FORWARD LOOKING STATEMENTS

Certain statements in this report relate to the future, including forward looking statements relating to the Company's financial position, strategy and expected operating results. These forward-looking statements involve known and unknown risks, uncertainties, assumptions and other important factors that could cause the actual results, performance or achievements of the Company to be materially different from future results, performance or achievements expressed or implied by such statements. Actual events or results may differ materially from the events or results expressed or implied in any forward-looking statement and deviations are both normal and to be expected. Other than required by law, neither the Company, their officers nor any other person gives any representation, assurance or guarantee that the occurrence of the events expressed or implied in any forward-looking statements will actually occur. You are cautioned not to place undue reliance on those statements.